

Direct Sets, Limits

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1 Introduction

A set J is said to be a directed set if $\leq$ on J is reflexive, transitive, all finite subsets of J have an upper $\leq$ -element.

Let $\mathcal{A} = \{A_\alpha\}_{\alpha \in J}$ be a collection of structures indexed by a directed set J , such that for all $\alpha \leq \beta$, we have homomorphisms $f_{\alpha\beta} : X_\alpha \rightarrow X_\beta$ st

$$(1) f_{\alpha\alpha} = \text{id}_{A_\alpha},$$

$$(2) \text{ If } \alpha \leq \beta, \beta \leq \gamma, \text{ then } f_{\beta\gamma} \circ f_{\alpha\beta} = f_{\alpha\gamma}.$$

If $\alpha \leq \beta$ and $\beta \leq \alpha$, then $A_\alpha \cong A_\beta$. $f_{\beta\alpha} \circ f_{\alpha\beta} = f_{\alpha\alpha} = \text{id}_{A_\alpha}$ and $f_{\alpha\beta} \circ f_{\beta\alpha} = f_{\beta\beta} = \text{id}_{A_\beta}$, so $f_{\alpha\beta}$ is bijective, i.e., A_α, A_β are isomorphic.

We can take the disjoint $\bigsqcup_{\alpha \in J} A_\alpha$ of $\mathcal{A}$. Identify $x \in A_\alpha$ with $\{x, A_\alpha\}$. If $\{x, A_\alpha\} = \{y, A_\beta\}$, then $x = y$ or $x = A_\beta, y = A_\alpha$, so $y \in x$ and $x \in y$, cont Regularity. Further, to force $\alpha = \beta$, identify as coordinate points the element $x \in A_\alpha$ by $\{\{\alpha\}, \{\alpha, \{x, A_\alpha\}\}\}$.

If we have a countable chain of structures $\{A_n\}_{n < \omega}$ such that there is a homomorphism $f_n : A_n \rightarrow A_{n+1}$, f_{nm} is determined by $f_{nn} = \text{id}_{A_n}$ and for $n < m$, $f_{nm} = f_{m-1} \circ \dots \circ f_{n+1} \circ f_n : A_n \rightarrow A_m$.

Define the ER $=$ on A st $x = y$ if $x \in A_\alpha, y \in A_\beta, \alpha \leq \beta, \beta \leq \alpha$, and $f_{\alpha\beta}(x) = y$. $=$ is reflexive, symmetric, and transitive.

Sps $x \in A_\alpha, y \in A_\beta$ and let $\gamma \geq \alpha, \beta$. Is it true $f_{\alpha\gamma}(x) = f_{\beta\gamma}(y)$ (in the strict sense of equality, and therefore $=$ as well). From $f_{\alpha\beta}(x) = y \rightarrow f_{\beta\gamma} \circ f_{\alpha\beta}(x) = f_{\beta\gamma}(y) \rightarrow f_{\alpha\gamma}(x) = f_{\beta\gamma}(y)$.

Let $\sim$ be the following ER on A : Let $x \in A_\alpha, y \in A_\beta$. $x \sim y$ if there exists $\gamma \geq \alpha, \beta$ st $f_{\alpha\gamma}(x) = f_{\beta\gamma}(y)$.

If $x_\alpha \sim y_\beta$, then for any $\delta \geq \alpha, \beta$, $f_{\alpha\delta}(x) \sim f_{\beta\delta}(y)$. Pick $\epsilon \geq \delta, \gamma$. Then

$$f_{\alpha\epsilon}(x) = f_{\gamma\epsilon} \circ f_{\alpha\gamma}(x) = f_{\gamma\epsilon} \circ f_{\beta\gamma}(y) = f_{\beta\epsilon}(y).$$

Let $\mathcal{A}$ be a collection of groups. Suppose $x_\alpha \stackrel{\beta}{\sim} x_{\alpha'}$ and y_γ . Define $x \cdot y = f_{\alpha\delta}(x) \cdot f_{\gamma\delta}(y)$ for some $\delta \geq \alpha, \gamma$.